

PSI

Habitat imaging analysis of local failures in skull-base cancer patients treated with proton therapy

MSc Semester Project

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Predicting Local Failure (LF)

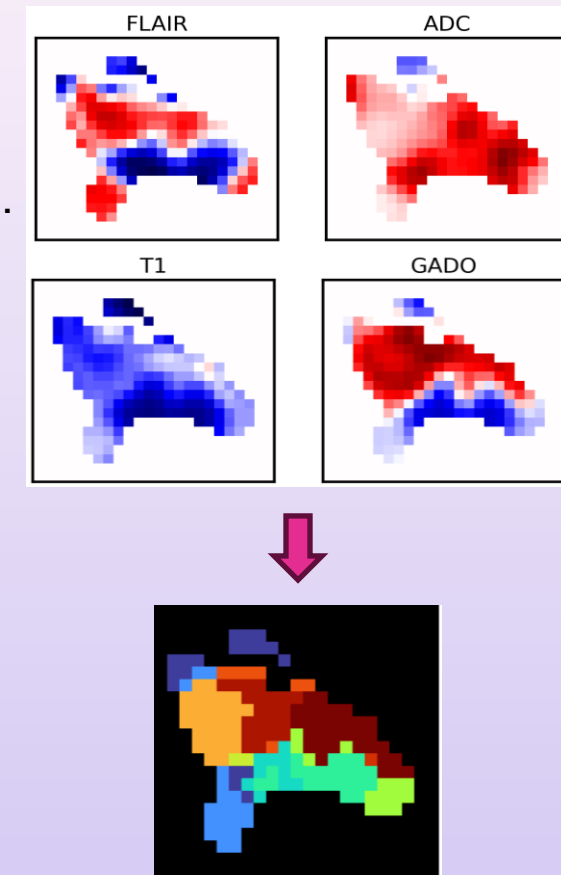
- LF still occurs in some skull-base cancer patients despite accurate PT.
- Conventional TCP modeling ignores **spatial tumour heterogeneity** – purely dose-based.
- Multi-modal MRI images can provide complementary information.
- Combination of images for voxel-wise characterization: **Habitats**

Hypothesis

- Habitat formation can capture spatial intra-tumour heterogeneity.
- Habitat features from pre-RT images may improve outcome prediction.

Goal

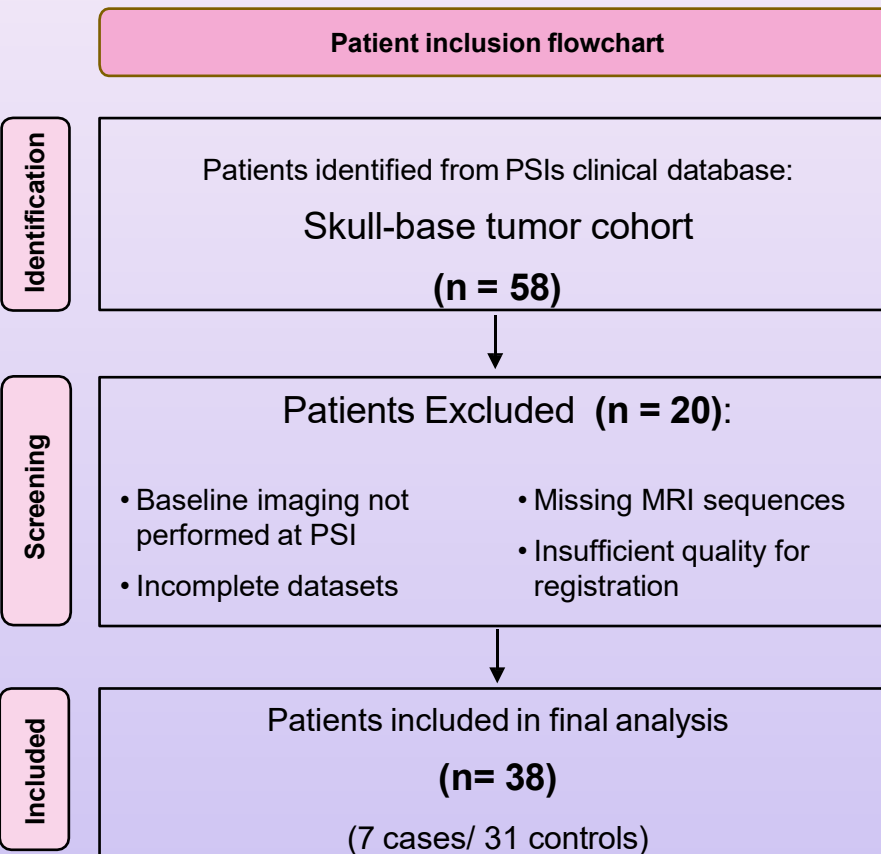
Assess the significance of habitat imaging analysis for predicting local failure after proton therapy



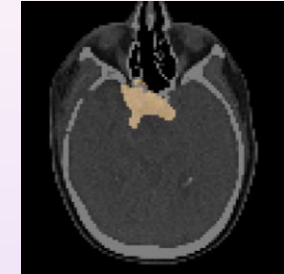
Study context

Clinical Cohort

- Skull-base cancer patients treated with proton therapy at PSI



Planning CT



Baseline MRI Images:

- T1-weighted
- FLAIR
- Gadolinium-enhanced T1-weighted (GADO-T1)
- ADC (DWI)

Clinical Data:

- Contour of LF and time of event recorded
- **NO dosimetry** information considered

My project: Habitat generation pipeline

Existing Radiomics Pipeline:

Registration

Masking

Intensity
Normalization

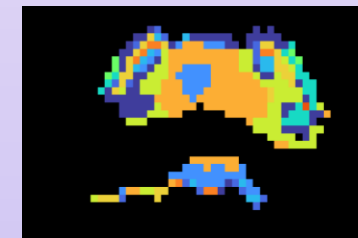
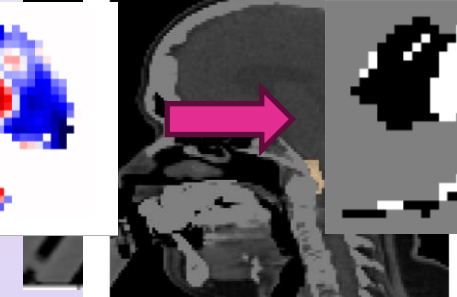
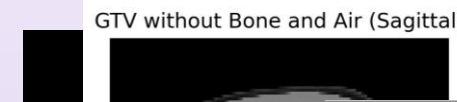
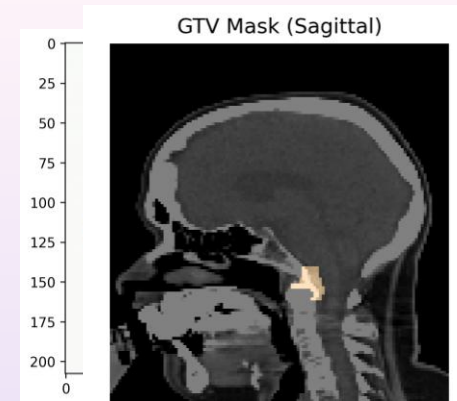
Feature Extraction

Survival Modeling

Soft tissue
extraction
within GTV

Intensity
Clustering

Habitat
Formation



deviation

$$\frac{\sigma - I_{\text{median}}}{AD}$$

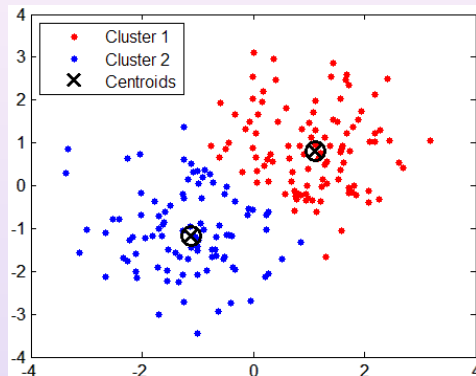
Habitat construction

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16
ADC	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
FLAIR	L	L	H	H	L	L	H	H	L	L	H	H	L	L	H	H
T1	L	L	L	L	H	H	H	H	L	L	L	L	H	H	H	H
GADO	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H



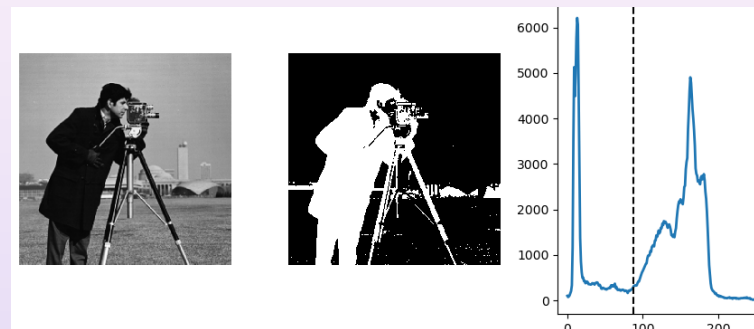
Clustering techniques

KMeans



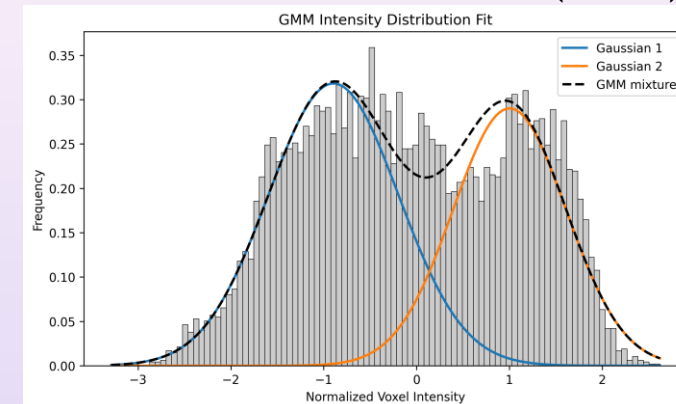
<https://dzone.com/articles/10-interesting-use-cases-for-the-k-means-algorithm>

Otsu Thresholding



https://scipy-lectures.org/packages/scikit-image/auto_examples/plot_threshold.html

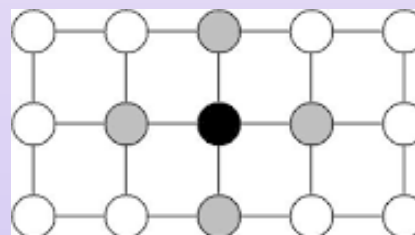
Gaussian Mixture Model (GMM)



GMM with Spatial Coordinates

$$[I, \sigma_x x, \sigma_y y, \sigma_z z]$$

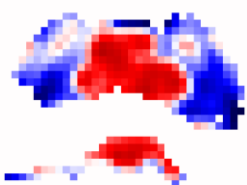
GMM/Spatial GMM + Markov Random Field (MRF) Smoothing



$$E = E_{\text{data}} + E_{\text{smooth}}$$

$$E_{\text{smooth}} = \lambda (N - n_x)$$

Normalized FLAIR



KMeans



Otsu



GMM



Spatial GMM



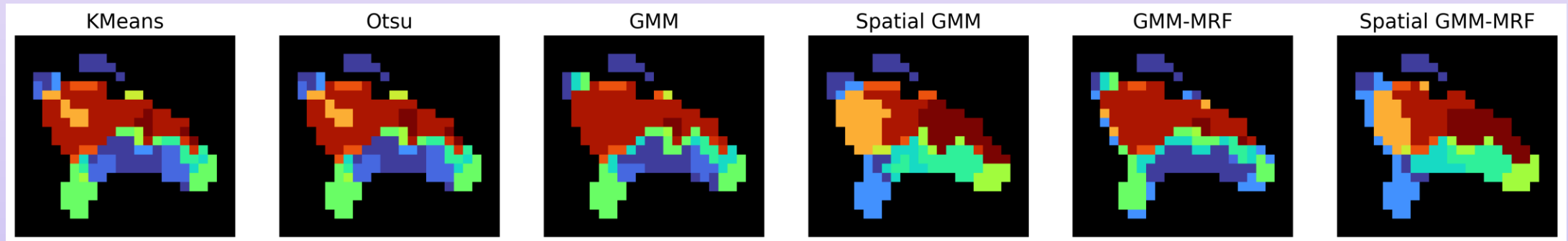
GMM-MRF



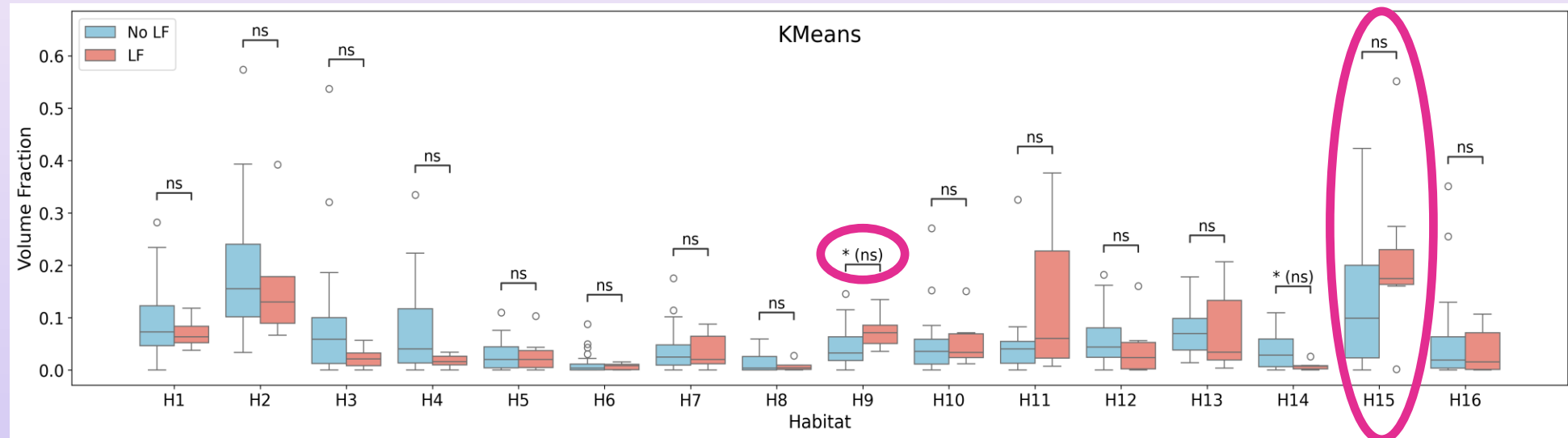
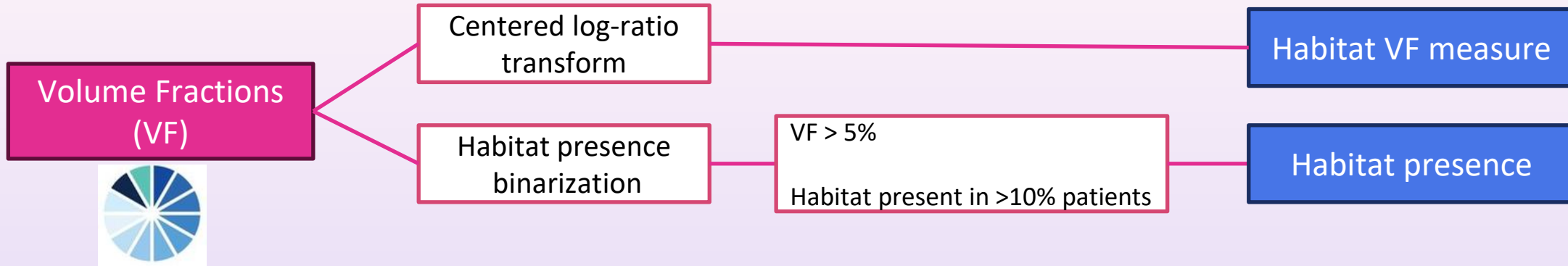
Spatial GMM-MRF



Clustering Method	Description	Constraints
KMeans	Minimize variance within each clusters	Intensity
Otsu	Optimal threshold that maximizes variance between clusters	Intensity
GMM	Probabilistic gaussian clustering	Intensity
Spatial GMM	Includes spatial coordinates as variables	Intensity + Spatial
GMM-MRF	Neighbourhood-based spatial smoothing	Intensity + Spatial
Spatial GMM-MRF	Includes coordinates and spatial smoothing	Intensity + Spatial



Habitat analysis between patient groups



- Mann–Whitney U-tests with p-value correction for multiple comparisons (Benjamini Hochberg).
- **No** statistically significant differences between LF and no LF - limited sample size and outliers.
- Visual distribution differences were observed.

Survival analysis

Cox Proportional Hazards (CPH) Model

- Relates covariates (features) to the hazard function describing the **instantaneous risk** at a given time.
- Least absolute shrinkage and selection operator (LASSO) penalty used - fewer and more robust selection

$$h(t|x) = h_0(t)e^{\beta^T x}$$

$$\mathcal{L}_{\text{penalized}} = -\ell(\beta) + \alpha \sum_{j=1}^p |\beta_j|$$

Habitat Volume Fractions

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16
KMeans			-0.20	-0.05					0.60			-0.05				
Otsu			-0.28						0.41	0.05	0.12	-0.21		-0.06		
GMM			-0.22	-0.15					0.45			-0.11				
GMM-MRF									0.08							
Spatial GMM				-0.20				-0.05		0.09						
Spatial GMM-MRF				-0.07				-0.14								

- H9** (GADO-T1) gets selected as the **hazard-increasing** habitat.
- H3** (FLAIR) and **H4** (FLAIR & ADC) show **hazard-decreasing** effects.

Habitat Presence

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16
KMeans				-0.97					0.58							
Otsu				-0.96					0.56							
GMM			-0.34													
GMM-MRF		0.61		-0.59			0.41		1.16						0.09	
Spatial GMM				-0.28			-0.91			0.29						
Spatial GMM-MRF		0.74		-0.31			-0.28		0.35	0.45						

- NO** information about **robustness** of habitat selection or predictability **performance** of each method.

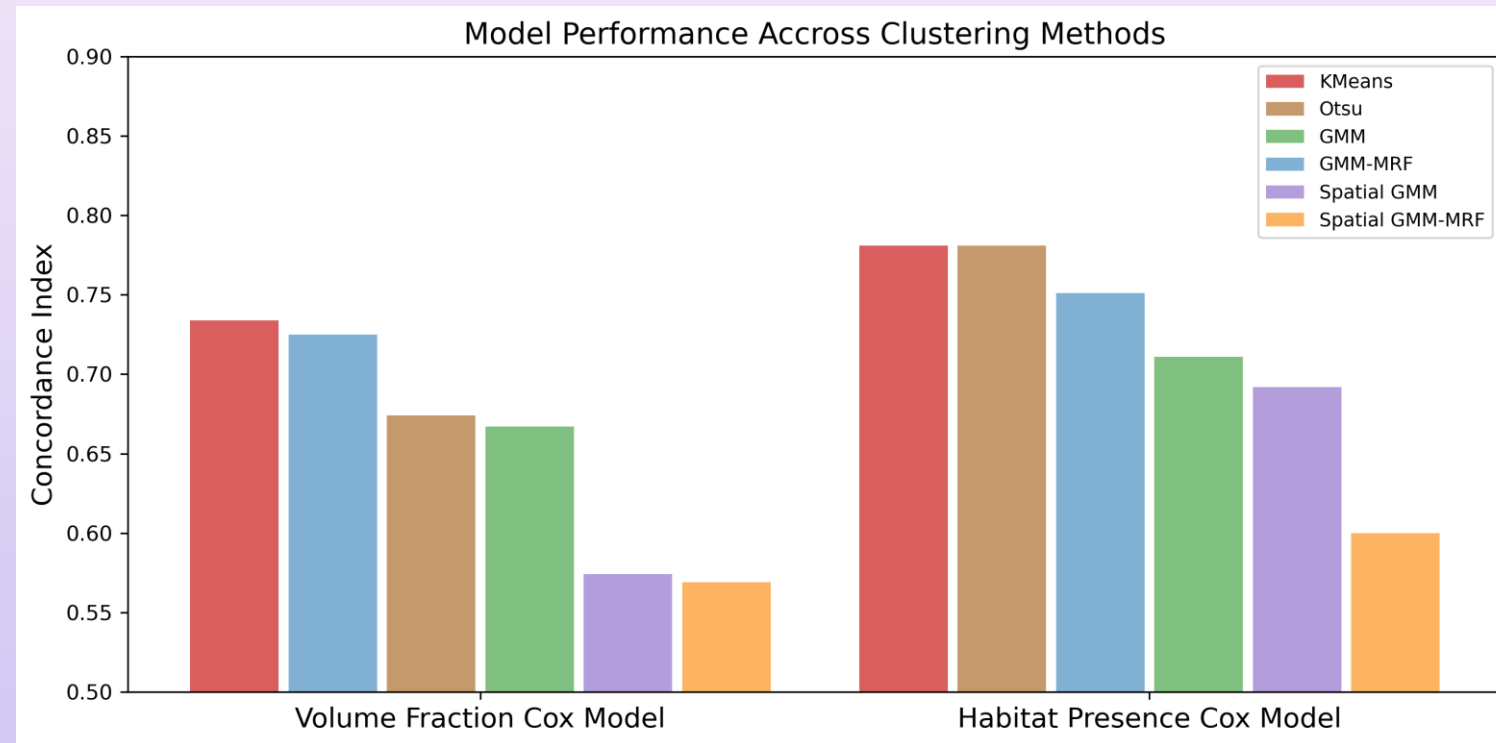
Model performance - Concordance index



Evaluates how well the model ranks patients according to LF risk over time.

- Proportion of correctly ranked patients

- **KMeans** achieved higher overall predictive performance.
- **Spatial GMM** and **Spatial GMM-MRF** obtained the lowest predictive index.
- **Intensity-based** approaches demonstrated more robust prognostic performance than **spatially constrained** methods.
- **Habitat presence** models obtained higher predictive indexes than the corresponding volume fraction models across clustering methods.



Model performance - Stability analysis

Bootstrap Stability Analysis

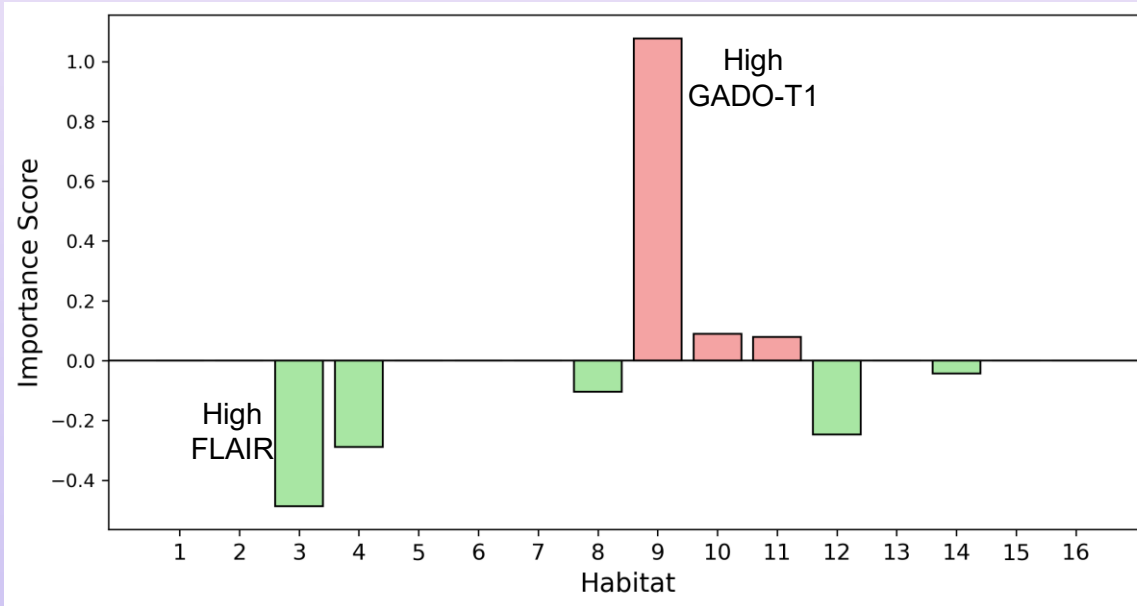
- 2000 bootstraps
- 80% proportion (5 LF, 24 no LF)



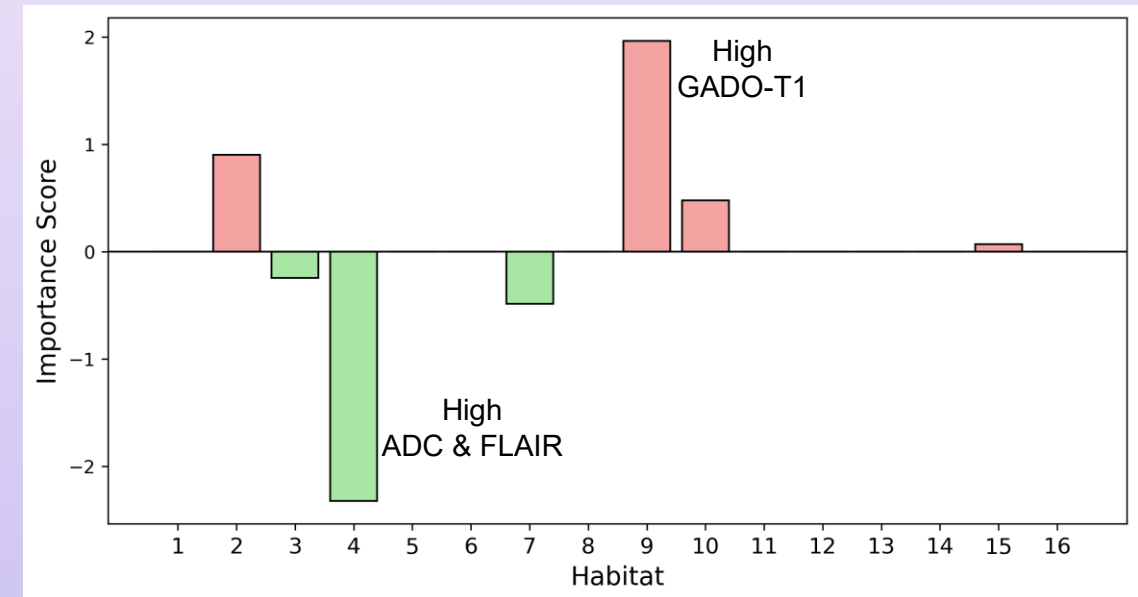
Stability Coefficient
% of bootstraps selected

$$I_h = \sum_{m=1}^M (\beta_{h,m} \cdot s_{h,m} \cdot C_m)$$

Habitat Volume Fractions



Habitat Presence



Methodology

Developed a multi-modal MRI habitat generation pipeline to characterize intra-tumour heterogeneity and predict LF after proton therapy.

Clinical Hints

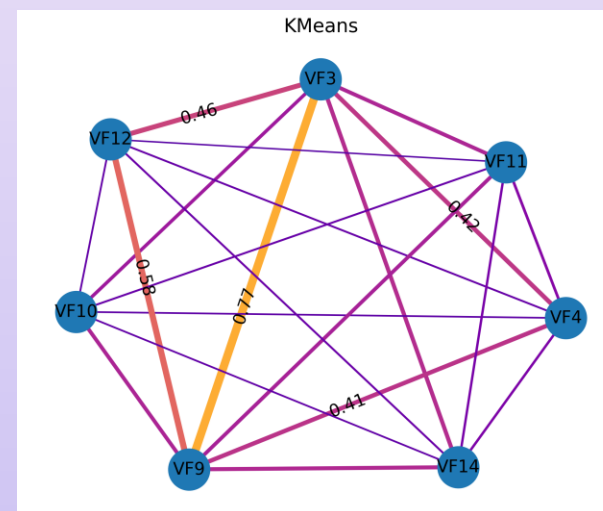
High GADO-T1 and high ADC/FLAIR habitats emerged as potential predictors of LF.

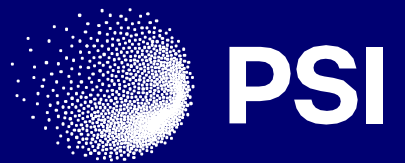
Limitations

No dosimetry information considered.
Several MRI images required – not always available.

Future work

Investigation of spatial correspondence between habitats.
Analysis of habitat interactions.





Thank you!

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